Modeling Volcanic Flows

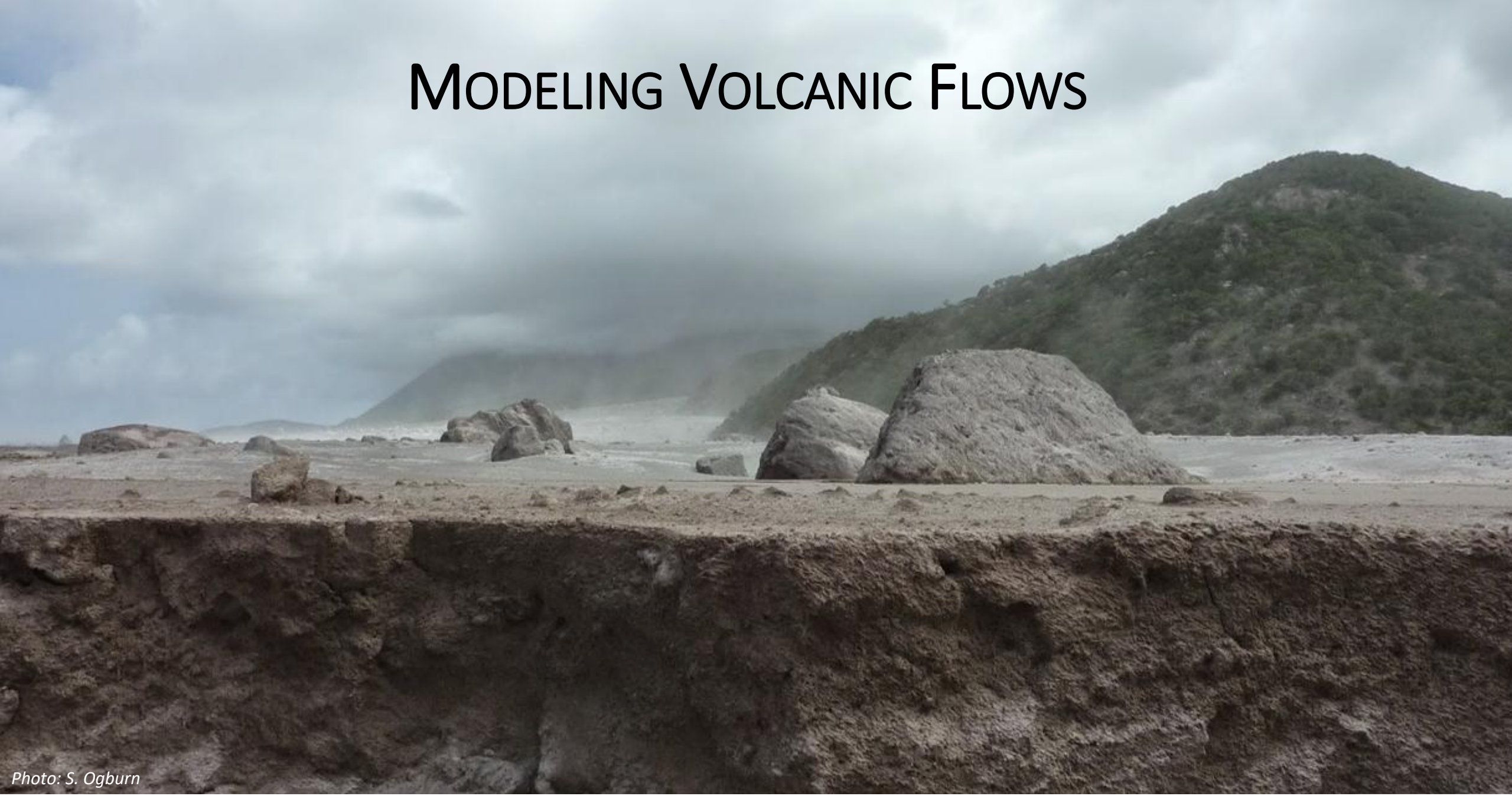
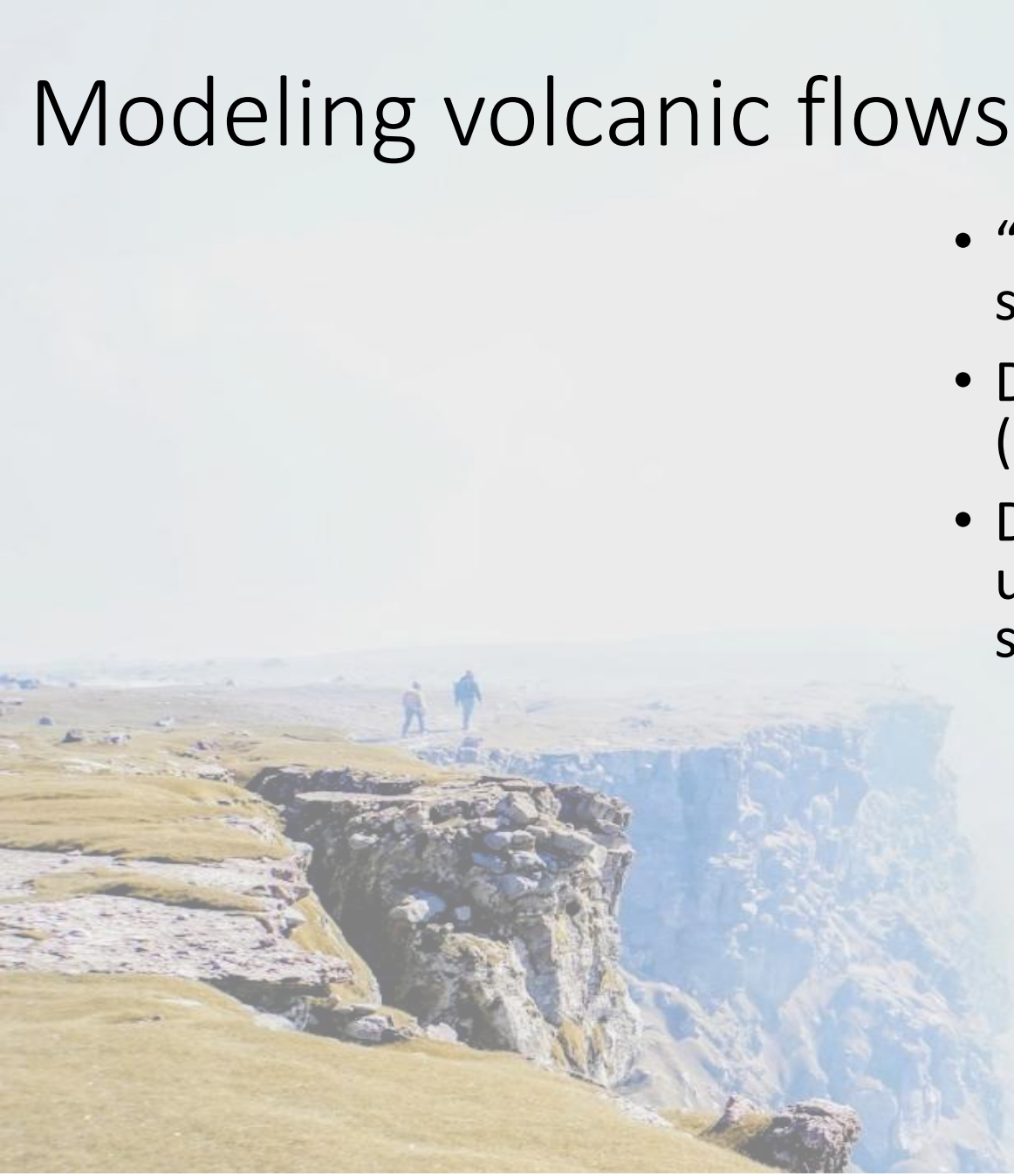


Photo: S. Ogburn

VDAP

USGS

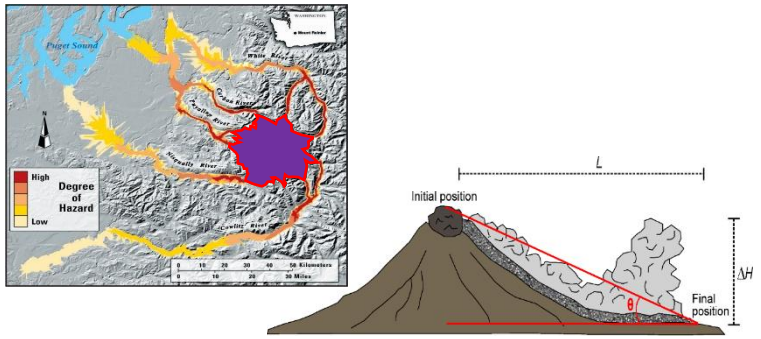
Modeling volcanic flows



- "All models are wrong, but some are useful" – statistician George Box
- Different models are wrong in different ways (make different simplifications and assumptions)
- Different models might be more wrong (or more useful) for certain questions, types of hazards, or situations
  - One model might be better for simulating large PDCs than small PDCs, or one model might be better at simulating velocities while another is better at simulating inundation
  - One model might be better if you only have a short amount of time, or if you have a poor quality DEM

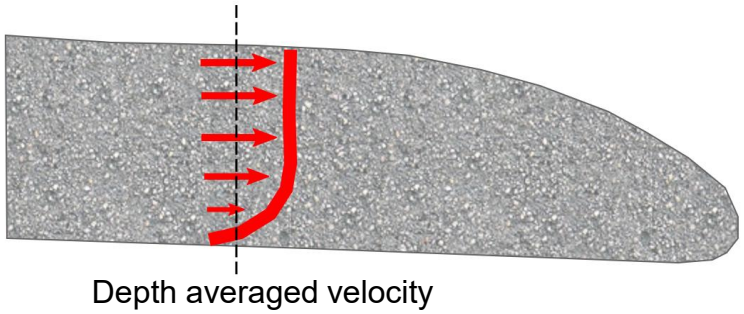
Increasing complexity and computational time

Statistics-based (Empirical)

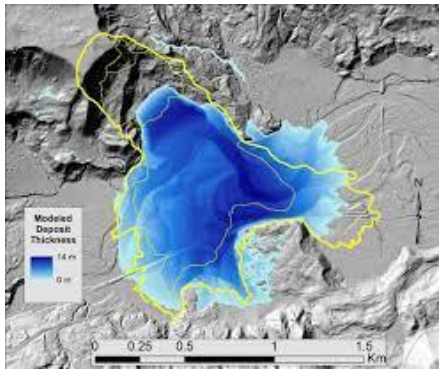


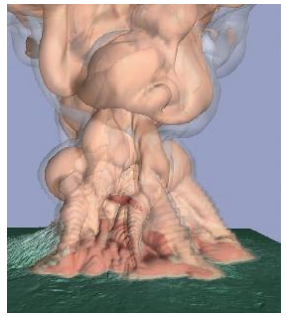
Empirical models
-Based on local and global data
-Runout or inundated areas derived from statistical relationships with flow volume
-Ex: Energy cone (Malin & Sheridan, 1982), FLOW2D (Sheridan & Macias, 1992), LAHARZ (Iverson et al. 1998, Schilling et al. 2014)

Physics-based



Depth-averaged models
-Relatively thin, dense, concentrated flows
-Frictional interaction at the bed or constant retarding shear stress controls flow motion
-Ex: TITAN2D (Patra et al. 2005), VolcFlow (Kelfoun & Druitt , 2005)





Multiphase models
-Separate conservation equations for multiple particle types
-Drag between the gas and particles transmits momentum between phases
-Ex: PDAC (Ongaro et al. 2008), (Dufek & Bergantz 2007), DCLAW (George et al. 2014

Statistical/empirical models

- Because they are based on global data and they have good *predictive power* (what you need for a hazard map)
- Rapid assessment and portrayal of hazard zones
- Especially good for broad regions (such as an entire volcano) where considerable uncertainty exists about the locations and sizes of future flows.
- Can run statistical models for the best-fit regression and the upper and lower confidence bounds to show the range of uncertainty to reflect the effects of natural variability
- Simplicity means they can't answer questions about flow behavior (low *explanatory power*)
- LAHARZ, energy cones

Physical models

- Great *explanatory power*, which makes them good for detailed hazard assessments or for answering specific questions.
- But they also have more input parameters—and more sources of uncertainty—than a simple statistical models.
- In order to use for hazard maps, you can use them for scenario mapping or run them probabilistically.
- Can take a long time to run, especially probabilistically.
- Physical models with fewer input parameters are easier to run probabilistically than those with many input parameters.
- DCLAW, TITAN2D, VolcFlow

An example of different models

LAHARZ (statistical)

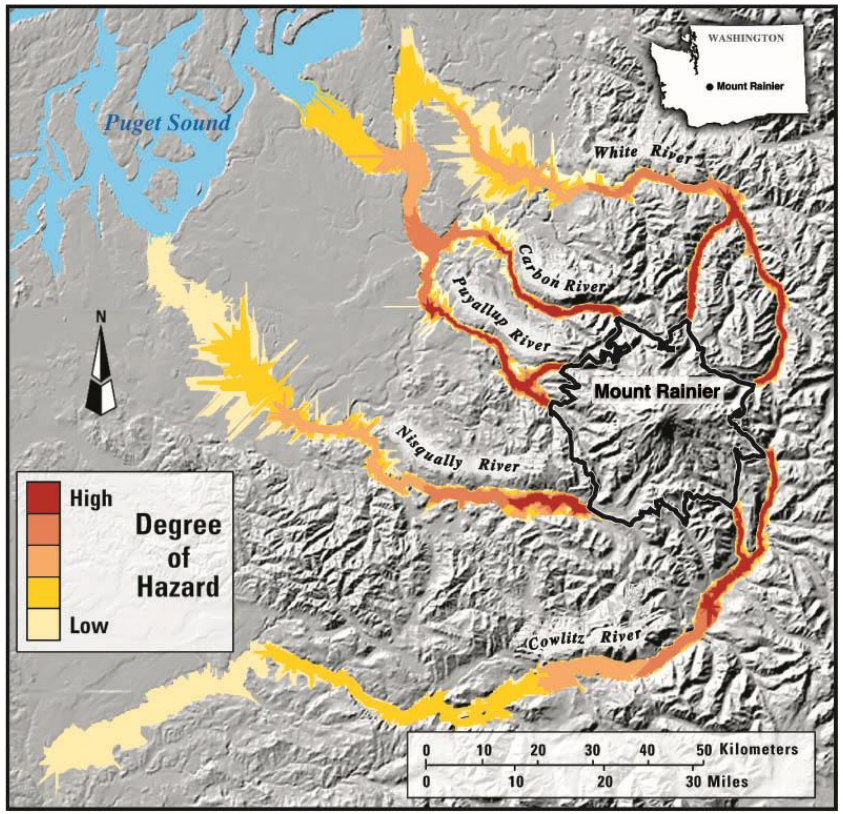
- Empirically based
- Few model input parameters (DEM, stream network, start locations, volumes)
- Vary volumes
- Gives simple maps of probable lahar inundation for a given volume
- Quick, not much computing power

D-CLAW (physical)

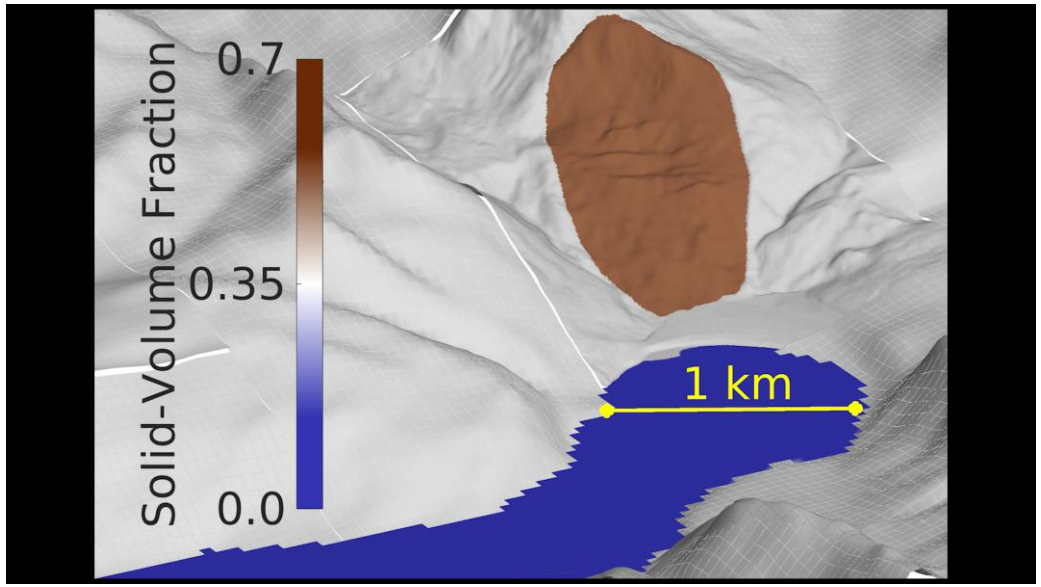
- Physical flow model
- Many model input parameters (precise 3D source area, 8 different material properties)
- Input parameters are very difficult to know ahead of time so must be varied with many simulations
- Simulates dynamics of flows and gives inundation, velocity, pore pressure evolution
- Computationally demanding, hard to run probabilistically

An example of different models

LAHARZ (statistical)



D-CLAW (physical)



An example of different models

Energy Cone (statistical)

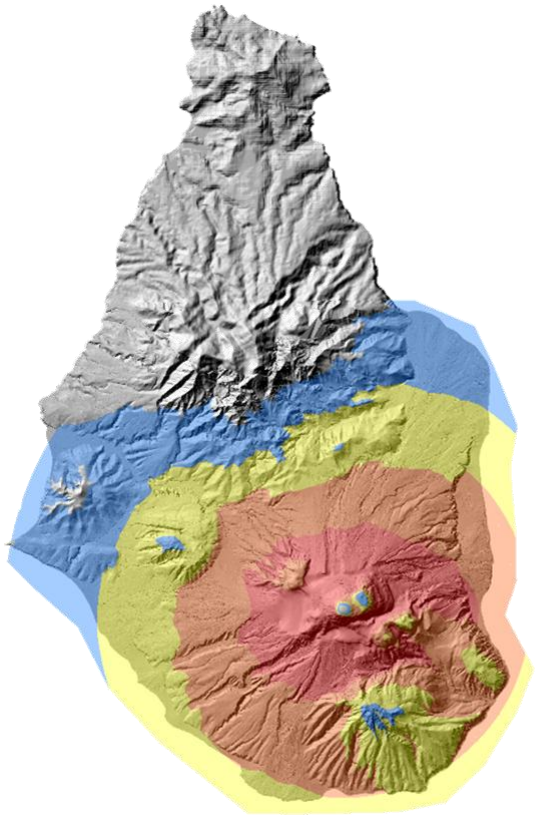
- Empirically based
- Few model input parameters (regression equations from global data relate volume to H/L)
- Vary volumes
- Gives simple maps of PDC inundation for a given volume
- Quick, not much computing power

TITAN2D (physical)

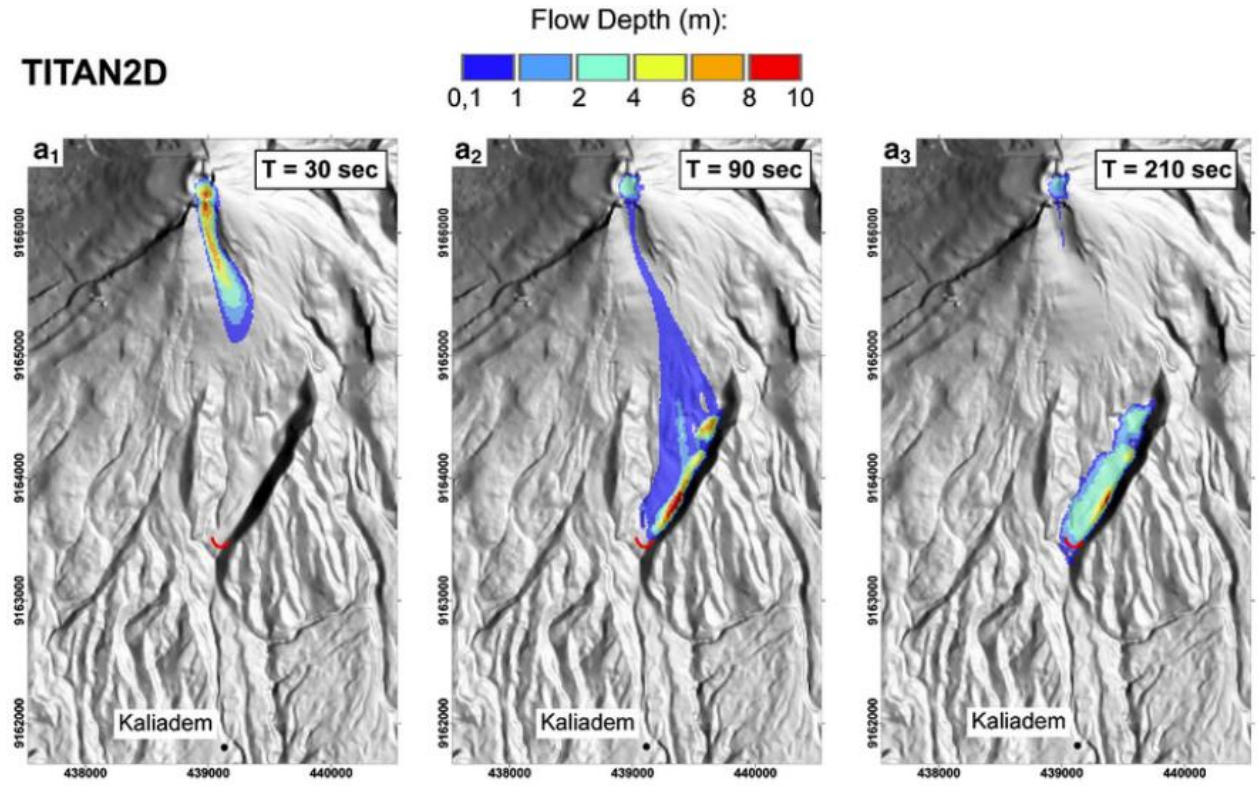
- Physical flow model
- More model input parameters (3-5 parameters)
- Input parameters are can be difficult to know ahead of time so must be varied
- Simulates dynamics of flows and gives inundation, velocity, thickness
- Somewhat computationally demanding

An example of different models

Energy cone (statistical)



TITAN2D (physical)

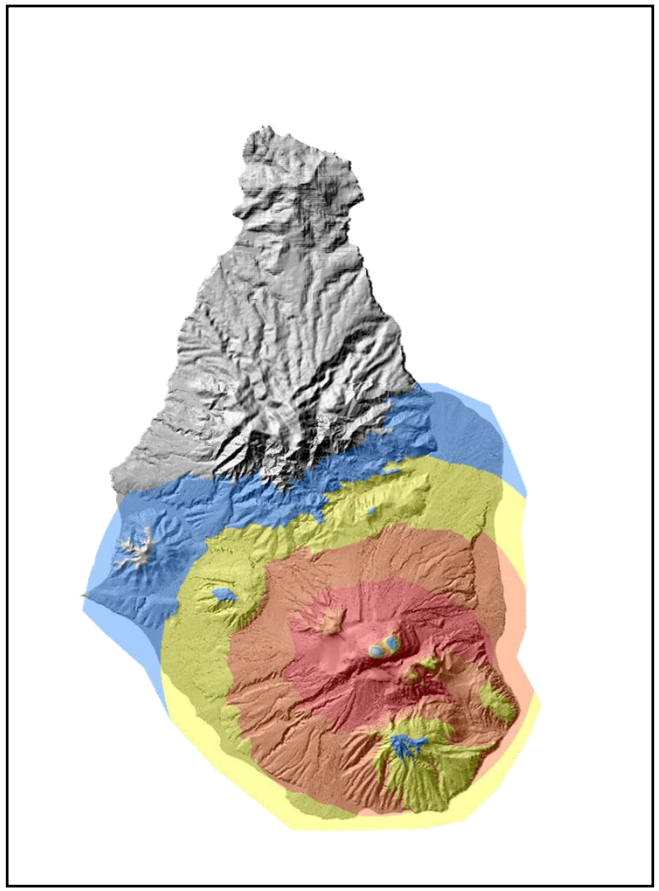


Scenario vs. Probabilistic modeling

Scenario-based or deterministic modeling	Probabilistic modeling
Model a few scenarios (Most-likely, worst-case, a set of volumes, a past well-know eruption)	Model many/all scenarios
Vary only a few parameters and use only a few parameter options (e.g., model 3 different volumes)	Vary more parameters randomly across the entire parameter probability distribution
Assumes that the eruption or event happens (e.g., conditional on eruption)	Can be conditional on eruption or not (e.g., could be probability of inundation given an eruption occurs vs. annual probability that considers how frequent eruptions happen)
Quick (a few model runs)	Slow (many, ~10,000, model runs)
May be easier to communicate to the public	Probabilities may be harder to communicate, but you could decide on thresholds to make simpler zones

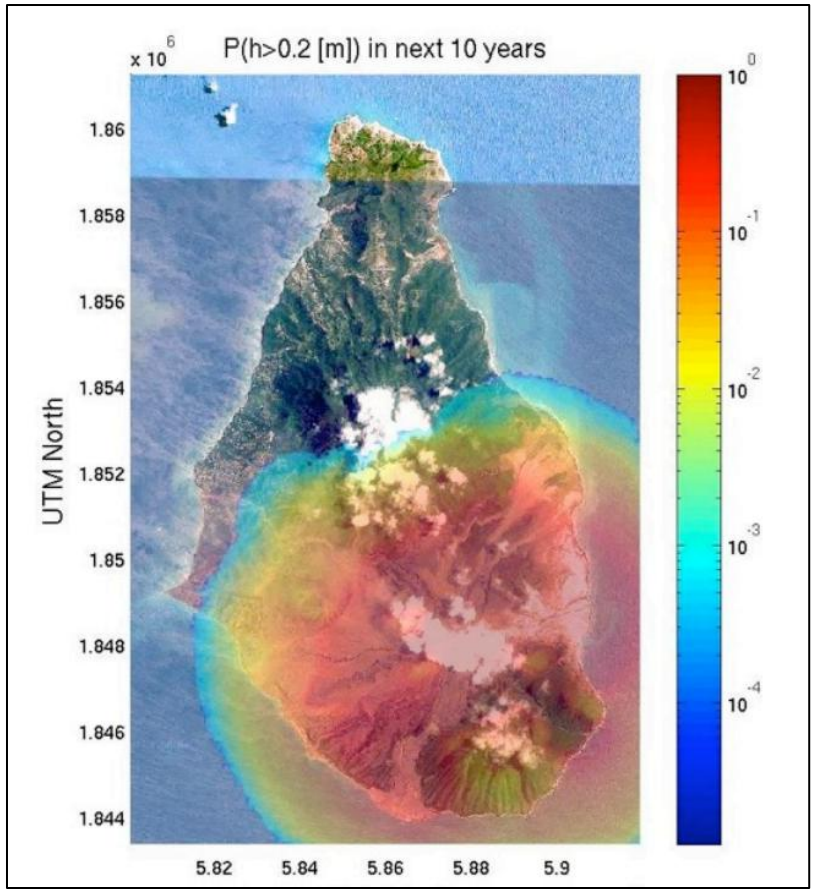
Note: geophysical models like TITAN2D can be run deterministically (scenarios) or probabilistically
Because of our time constraints we will only do scenario modeling!

Scenario modeling



4 ΔH/L energy cones
(for 0.0001 to 0.1 km^3)
< 5 minutes on 1 processer

Probabilistic modeling



100,000 TITAN2D model runs
(for 0.0001 to 0.1 km^3)
~1000 hours on 1024 processers

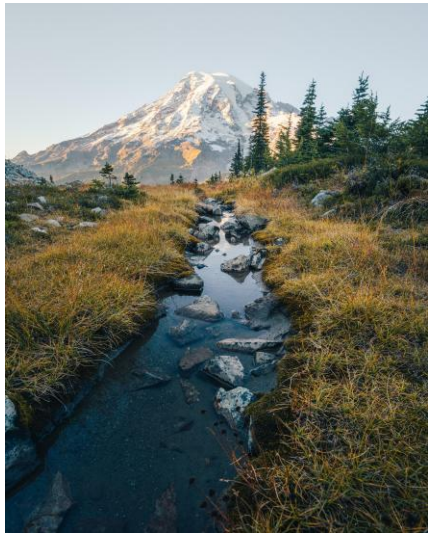
Using models for hazard mapping

ONE MODEL RUN ≠ HAZARD MAP

- For hazard mapping, the input parameters ***should be varied***
- **For LAHARZ:** use a variety of volumes down different valleys
- **For energy cone:** use a variety of ΔH/L values to show hazard from different PDC volumes
- **For TITAN2D/VolcFlow/other physical models:** use different volumes, different basal frictions angles, different pile geometries, and different directions (fully probabilistic hazard maps require 1000's of runs using a sampling of input parameter distributions)
- Compare multiple models and consider geologic evidence of past flows
- There's lots of uncertainty in modeling PDCs, especially column-collapse flows!

Models are tools

Stream networks and topography



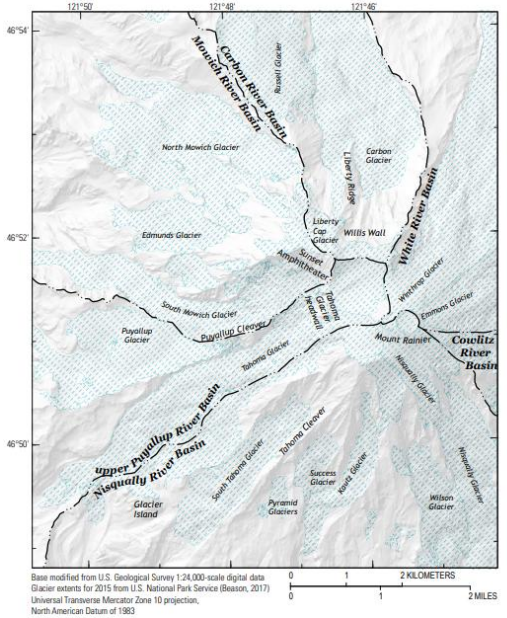
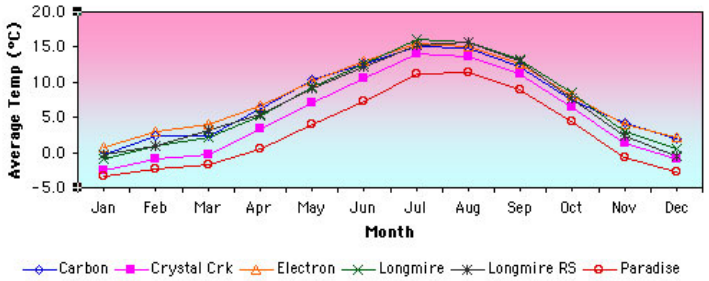
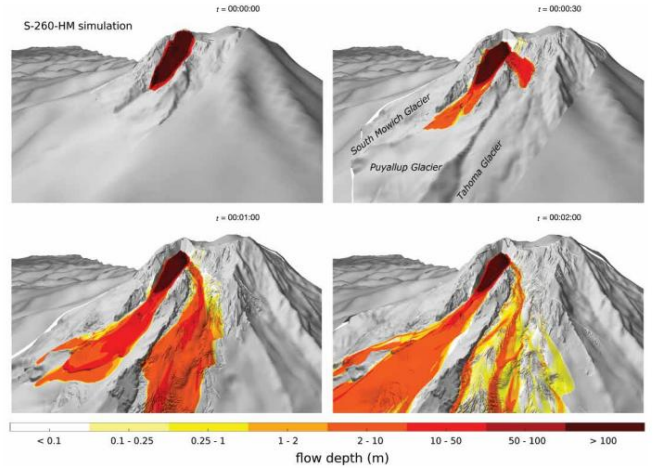


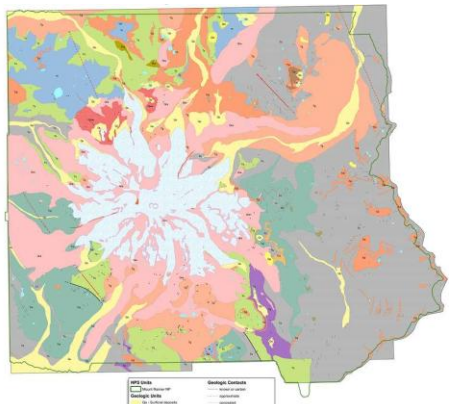
Figure 7. Shaded relief map of Mount Rainier showing the locations of river basin divides and glaciers below the Sunset Amphitheater and the Tahoma Glacier headwall. Note that the divide between the Puyallup and Nisqually River basins runs the length of Tahoma Glacier in between the more prominent topographic divides formed by the Puyallup Cleaver and the Tahoma Cleaver.

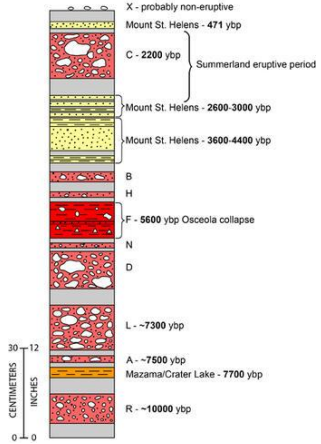


Current conditions

Other models







Geology

Process knowledge

